

1. Borough Trends in Complaints and Violations

We used bar charts and comparative visuals to highlight borough-wise disparities in complaint volumes and violations. Brooklyn emerged as the borough with the highest number of complaints, while the Bronx had the highest percentage of violations (46%). These graphs effectively illustrate the severity of housing conditions in certain areas and set the stage for deeper analysis.

2. Complaint Types Breakdown

A categorical bar chart was used to display the most common housing issues, such as heat/hot water complaints dominating in Brooklyn and the Bronx. This choice allowed us to pinpoint which problems require urgent intervention. A stacked bar or grouped bar chart was ideal here, as it shows borough-wise distributions clearly while allowing easy comparisons.

3. Complaints Leading to Violations

To assess the effectiveness of NYC's housing enforcement, we visualized the ratio of complaints that result in violations. A proportional bar chart effectively captured this, demonstrating that while Brooklyn had the highest complaint volume, the Bronx had the most severe enforcement outcomes. This reinforced the need for targeted action in high-risk areas.

4. Resolution Timelines and Efficiency

A combination of line charts and bar charts was used to analyze complaint resolution times. These graphs illustrated that Queens and Staten Island had the fastest resolution times, while the Bronx experienced significant delays. The choice of a time-series visualization here allowed us to highlight borough-level inefficiencies over time.

5. Geographic Hotspots for Complaints

A heatmap or spatial analysis visualization was included to identify areas with the most complaints. This was crucial in prioritizing resource allocation, as geographic insights help city officials target high-risk neighborhoods.

6. Trends Over Time

To track complaint trends across years, we used a time-series graph. This choice helped reveal patterns, such as the spike in water leak complaints in 2022 and fluctuations in heat/hot water complaints. Without this visualization, it would be difficult to identify whether certain problems were improving or worsening.

7. Monthly Trends in Complaints

A seasonal trend analysis was necessary to determine fluctuations in complaint types. A line chart was the best way to showcase how heat/hot water complaints peak in December and January, reinforcing the need for proactive measures during colder months.

Conclusion

The final graphs were chosen based on clarity, insightfulness, and relevance to key findings. Each visualization played a critical role in highlighting borough disparities, complaint trends, violation resolutions, and seasonal patterns.